

Clinical Protocol for Type 2 Diabetes Mellitus Management

1. Diagnostic Thresholds

Type 2 Diabetes Mellitus (T2DM) is diagnosed based on any of the following criteria:

- HbA1c $\geq$ 6.5%.
- Fasting Plasma Glucose (FPG) $\geq$ 126 mg/dL (7.0 mmol/L).
- 2-hour oral glucose tolerance test (OGTT) value $\geq$ 200 mg/dL.
- Random plasma glucose $\geq$ 200 mg/dL in a patient with classic symptoms of hyperglycemia (polyuria, polydipsia, unexplained weight loss).

2. Treatment Algorithm

First-Line Therapy: Metformin (initial dose 500 mg once or twice daily, titrate up to 2000 mg daily as tolerated). Metformin should be continued unless contraindicated (e.g., eGFR $<$ 30 mL/min/1.73m²).

Combination Therapy (if HbA1c target not met after 3 months):

- Add SGLT2 Inhibitors (e.g., Empagliflozin, Dapagliflozin) for patients with heart failure or diabetic nephropathy.
- Add GLP-1 Receptor Agonists (e.g., Semaglutide, Liraglutide) for patients with high cardiovascular risk.
- Add DPP-4 Inhibitors (e.g., Sitagliptin) or Sulfonylureas (e.g., Glimepiride) as alternative cost-effective options.
- Insulin therapy should be initiated if HbA1c is $>$ 9.0% with severe symptoms.

3. Lifestyle and Monitoring

Lifestyle Recommendations:

- Carbohydrate-controlled diet focusing on complex carbohydrates and high-fiber foods.
- Moderate-intensity exercise (at least 150 minutes per week, no more than 2 consecutive days without exercise).
- Weight management target: 5-7% weight reduction for patients who are overweight or obese.

Monitoring Guidelines:

- Measure HbA1c every 3 months for patients not meeting targets; twice yearly for stable patients meeting targets.
- Annual screening for diabetic nephropathy (urine albumin-to-creatinine ratio), retinopathy (dilated eye exam), and diabetic neuropathy (foot sensory examination).